

Stony Brook University
Instructor: Alejandro Melo Ponce

Spring 2017

MIDTERM EXAM II

NAME: _____

ID: _____

Instructions: Answer all questions. The total number of points for the exam is 100. The number of points assigned to each section is indicated. Calculators are allowed but other electronic devices or outside materials are NOT allowed. For questions with a numerical answer, give a numerical answer. For example

$$2 + V[X]$$

is NOT the final answer if you can compute a numerical value for $V[X]$.

At the back of the exam you are provided with a normal distribution table. You are required to compute probabilities using the normal distribution when asked to. This means that your answer **cannot** be something like

$$1 - \Phi(1.67),$$

you have to calculate that this number is actually 0.04746.

You are **NOT** allowed to leave the room until you submit your exam.

GOOD LUCK!

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Part I: Multiple Choice Questions (20 points)

1. When the estimated slope coefficient in the simple regression model, $\hat{\beta}_1$, is zero, then
 - (a) $R^2 = 0$
 - (b) $0 < R^2 < 1$.
 - (c) $R^2 = \bar{Y}$.
 - (d) $R^2 > \frac{SSR}{TSS}$
2. The following are all least squares assumptions with the exception of:
 - (a) The explanatory variable in the regression model is normally distributed.
 - (b) The conditional distribution of u_i given X_i has a mean of zero.
 - (c) Large outliers are unlikely.
 - (d) $(X_i, Y_i), i = 1, \dots, n$ are independently and identically distributed.
3. The OLS estimator is derived by
 - (a) minimizing the sum of absolute residuals.
 - (b) minimizing the sum of squared residuals.
 - (c) making sure that the standard error of the regression equals the standard error of the slope estimator.
 - (d) connecting the Y_i corresponding to the lowest X_i observation with the Y_i corresponding to the highest X_i observation.
4. In the simple linear regression model, the regression slope
 - (a) when multiplied with the explanatory variable will give you the predicted Y .
 - (b) indicates by how many units Y increases, given a one unit increase in X .
 - (c) represents the elasticity of Y on X .
 - (d) indicates by how many percent Y increases, given a one percent increase in X .
5. The slope estimator, $\hat{\beta}_1$, has a smaller standard error, other things equal, if
 - (a) the sample size is smaller.
 - (b) there is more variation in the explanatory variable, X .
 - (c) the intercept, β_0 , is small.
 - (d) there is a large variance in the error term, u .
6. The t -statistic is calculated by dividing
 - (a) the slope by the standard deviation of the explanatory variable.
 - (b) the slope by 1.96.
 - (c) the estimator minus its hypothesized value by the standard error of the estimator.
 - (d) the OLS estimator by its standard error.

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7. If the absolute value of your calculated t -statistic exceeds the critical value from the standard normal distribution, you can
- (a) safely assume that your regression results are significant.
 - (b) conclude that most of the actual values are very close to the regression line.
 - (c) reject the null hypothesis.
 - (d) reject the assumption that the error terms are homoskedastic.
8. Finding a small value of the p -value (e.g. less than 5%)
- (a) will only happen roughly one in twenty samples.
 - (b) indicates evidence in against the null hypothesis.
 - (c) implies that the t -statistic is less than 1.96.
 - (d) indicates evidence in favor of the null hypothesis.
9. One of the following steps is not required as a step to test for the null hypothesis:
- (a) compute the p -value.
 - (b) test for the errors to be normally distributed.
 - (c) compute the standard error of $\hat{\beta}_1$
 - (d) compute the t -statistic.
10. You extract approximately 5,000 observations from the Current Population Survey (CPS) and estimate the following regression function:

$$\widehat{ahe} = \underset{(1.00)}{3.32} - \underset{(0.04)}{0.45} \times Age, \quad R^2 = 0.02, \quad SER = 8.66$$

where ahe is average hourly earnings, and Age is the individual's age. Given the specification, your 95% confidence interval for the effect of changing age by 5 years is approximately

- (a) cannot be determined given the information provided.
- (b) [\$2.32, \$4.32]
- (c) [\$1.35, \$5.30]
- (d) [\$1.96, \$2.54]

Part II: Open Problems (80 points)

Interpreting Regression Output (30 points)

Suppose you run the following regression of monthly wage in the U.S. on lagged U.S. wage (i.e., wage in the previous month) and get

Source	SS	df	MS	Number of obs = 153		
Model	520247.469	1	520247.469	F(3, 36) =11568.47		
Residual	6790.64354	151	44.9711492	Prob > F = 0.0000		
Total	52038.113	152	3467.356	R-squared = 0.9871		
				Adj R-squared = 0.9870		
				Root MSE = 6.7061		

wage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lagwage	.9887841	.0091931	107.56	0.000	.3157855	1.006948
_cons	7.298613	5.050625	1.45	0.000	-2.680407	17.27763

1. If the wage in the U.S. was 8.00 in May 2006, what is the predicted wage for June 2006?
2. Draw a scatter-plot representing your sample. Plot also the estimated regression line. Explain why your graph looks the way it looks.
3. Without any calculation, are you more or less likely to reject the standard null hypothesis using a 90% confidence interval or a 95% confidence interval? Explain.

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Regression With a Binary Variable (30 points)

In an attempt to learn more about the determinants of heights in the US population, you collect some height data on 320,000 randomly selected males and 260,000 randomly selected females. After running *OLS* you have

$$\widehat{Height}_i = \frac{72}{(1.45)} - \frac{5}{(2.00)} \times Female, \quad R^2 = 0.64, SER = 8.2.$$

where Height is measured in inches and Female is a binary variable that equals one if the person is a female and 0 if the person is a male.

1. Interpret the estimate $\hat{\beta}_1 = -5$ of β_1 (i.e., the coefficient of the variable *Female*).
2. Would you reject the standard *two-sided* null hypothesis at the 5% significance level? Explain why.
3. Compute the mean height for men in the sample?

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Linear Regression Model (20 points)

1. Explain the difference between the population regression line and the OLS regression line.
2. State the three assumptions for the linear regression model.
3. Explain why under an *idealized randomized experiment* the OLS Assumption 1 holds (i.e., conditional independence of X and u holds).
4. What is the implication regarding the OLS estimator of the three OLS assumptions?

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Formulas

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad (1)$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}, \quad (2)$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}. \quad (3)$$

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \quad TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad SSR = \sum_{i=1}^n \hat{u}_i^2. \quad (4)$$

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS}. \quad (5)$$

$$SER = \sqrt{\frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2}. \quad (6)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n \hat{u}_i^2}. \quad (7)$$

$$\sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{V[(X_i - \mu_X)u_i]}{(V[X_i])^2} = \frac{1}{n} \frac{\sigma_{v_i}^2}{(\sigma_X^2)^2} \quad (8)$$

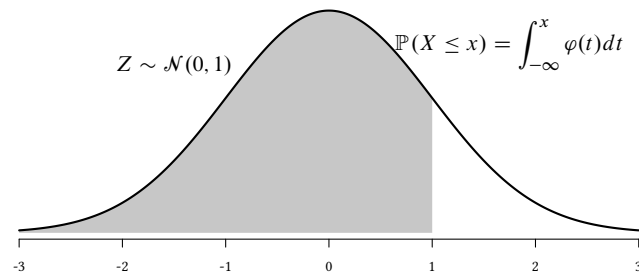
$$p\text{-value} = \Pr_{H_0} \left(|\hat{\beta}_1 - \beta_{1,0}| > |\hat{\beta}_1^{act} - \beta_{1,0}| \right) \text{ two-sided} \quad (9)$$

$$p\text{-value} = \Pr_{H_0} \left(\hat{\beta}_1 - \beta_{1,0} > \hat{\beta}_1^{act} - \beta_{1,0} \right) \text{ one-sided} \quad (10)$$

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)} \quad (11)$$

$$90\% \text{ confidence interval for } \beta_1 : \left[\hat{\beta}_1 \pm 1.64 \times SE(\hat{\beta}_1) \right]. \quad (12)$$

Cumulative Distribution of $\mathcal{N}(0, 1)$



	0.09	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01	0
-3	0.0010	0.0010	0.0011	0.0011	0.0011	0.0012	0.0012	0.0013	0.0013	0.0013
-2.9	0.0014	0.0014	0.0015	0.0015	0.0016	0.0016	0.0017	0.0018	0.0018	0.0019
-2.8	0.0019	0.0020	0.0021	0.0021	0.0022	0.0023	0.0023	0.0024	0.0025	0.0026
-2.7	0.0026	0.0027	0.0028	0.0029	0.0030	0.0031	0.0032	0.0033	0.0034	0.0035
-2.6	0.0036	0.0037	0.0038	0.0039	0.0040	0.0041	0.0043	0.0044	0.0045	0.0047
-2.5	0.0048	0.0049	0.0051	0.0052	0.0054	0.0055	0.0057	0.0059	0.0060	0.0062
-2.4	0.0064	0.0066	0.0068	0.0069	0.0071	0.0073	0.0075	0.0078	0.0080	0.0082
-2.3	0.0084	0.0087	0.0089	0.0091	0.0094	0.0096	0.0099	0.0102	0.0104	0.0107
-2.2	0.0110	0.0113	0.0116	0.0119	0.0122	0.0125	0.0129	0.0132	0.0136	0.0139
-2.1	0.0143	0.0146	0.0150	0.0154	0.0158	0.0162	0.0166	0.0170	0.0174	0.0179
-2	0.0183	0.0188	0.0192	0.0197	0.0202	0.0207	0.0212	0.0217	0.0222	0.0228
-1.9	0.0233	0.0239	0.0244	0.0250	0.0256	0.0262	0.0268	0.0274	0.0281	0.0287
-1.8	0.0294	0.0301	0.0307	0.0314	0.0322	0.0329	0.0336	0.0344	0.0351	0.0359
-1.7	0.0367	0.0375	0.0384	0.0392	0.0401	0.0409	0.0418	0.0427	0.0436	0.0446
-1.6	0.0455	0.0465	0.0475	0.0485	0.0495	0.0505	0.0516	0.0526	0.0537	0.0548
-1.5	0.0559	0.0571	0.0582	0.0594	0.0606	0.0618	0.0630	0.0643	0.0655	0.0668
-1.4	0.0681	0.0694	0.0708	0.0721	0.0735	0.0749	0.0764	0.0778	0.0793	0.0808
-1.3	0.0823	0.0838	0.0853	0.0869	0.0885	0.0901	0.0918	0.0934	0.0951	0.0968
-1.2	0.0985	0.1003	0.1020	0.1038	0.1056	0.1075	0.1093	0.1112	0.1131	0.1151
-1.1	0.1170	0.1190	0.1210	0.1230	0.1251	0.1271	0.1292	0.1314	0.1335	0.1357
-1	0.1379	0.1401	0.1423	0.1446	0.1469	0.1492	0.1515	0.1539	0.1562	0.1587
-0.9	0.1611	0.1635	0.1660	0.1685	0.1711	0.1736	0.1762	0.1788	0.1814	0.1841
-0.8	0.1867	0.1894	0.1922	0.1949	0.1977	0.2005	0.2033	0.2061	0.2090	0.2119
-0.7	0.2148	0.2177	0.2206	0.2236	0.2266	0.2296	0.2327	0.2358	0.2389	0.2420
-0.6	0.2451	0.2483	0.2514	0.2546	0.2578	0.2611	0.2643	0.2676	0.2709	0.2743
-0.5	0.2776	0.2810	0.2843	0.2877	0.2912	0.2946	0.2981	0.3015	0.3050	0.3085
-0.4	0.3121	0.3156	0.3192	0.3228	0.3264	0.3300	0.3336	0.3372	0.3409	0.3446
-0.3	0.3483	0.3520	0.3557	0.3594	0.3632	0.3669	0.3707	0.3745	0.3783	0.3821
-0.2	0.3859	0.3897	0.3936	0.3974	0.4013	0.4052	0.4090	0.4129	0.4168	0.4207
-0.1	0.4247	0.4286	0.4325	0.4364	0.4404	0.4443	0.4483	0.4522	0.4562	0.4602
0	0.4641	0.4681	0.4721	0.4761	0.4801	0.4840	0.4880	0.4920	0.4960	0.5000

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	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990